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CPT_S 422 Software Engineering Principles II

Homework 1

Questions

1. Briefly discuss how errors, faults, and failures are different. (4 points)

Errors are caused by programmers making mistakes. These errors can create faults in the code, also called bugs, which can then cause failures when the software does not behave as expected.

Questions 2 and 3 refer to the following method from an online store:

double shippingCost(int weight, int distance)

The shipping rate depends on which tier each input falls into:

Input	Unit	Valid Range	Tiers
Weight	Pounds	[1,50]	Light [1,10], Medium [11,30], Heavy [31,50]
Distance	Miles	[1,3000]	Regional [1,500], National [501, 3000]

Notes for Questions 2 and 3:

- List only the input values (weight, distance) for each test case. There is no need to provide

  expected outputs.
- Use the textbook (Jorgensen) definitions for all techniques.
- Use 25 as the nominal weight and 1500 as the nominal distance.
- Both parameters are int, so consider only integer inputs. Do not add classes for non-integer or null values.

2. Boundary value testing. (6 points)

a. List the normal boundary value test cases for this method, using the boundaries of each input's valid range.

(1, 1500)

(2, 1500)

(49, 1500)

(50, 1500)

(25, 1)

(25, 2)

(25, 2999)

(25, 3000)

(25, 1500)

b. How many test cases would robust, worst-case, and robust worst-case boundary value testing require? Show how you arrived at each number.

Normal Boundary Testing:

4n + 1

4(2) + 1 = 9

Robust Boundary Testing

6n + 1 (adds one below min and one above max)

6(2) + 1 = 13

Worst Case Boundary Testing

5^n

5^(2) = 25

Robust Worst Case Boundary Testing

7^n (adds one below min and one above max)

7^(2) = 49

c. Assuming each test case had the correct expected output, would your test cases from (a) catch an off-by-one error in the light/medium tier check, for example weight < 10 written instead of weight <= 10? Explain why or why not.

No. The normal boundary value test cases only test the boundaries of the overall valid input range, not the boundaries between shipping tiers. Since weight values 10 and 11 are not tested, an error such as weight < 10 instead of weight <= 10 may not be detected.

3. Equivalence class testing. (8 points)

a. Identify the valid and invalid equivalence classes for each input variable.

Weight

Invalid: weight < 1

Valid: 1-10 Light

Valid: 11-30 Medium

Valid: 31-50 Heavy

Invalid: weight > 50

Distance

Invalid: distance < 1

Valid: 1-500 Regional

Valid: 501-3000 National

Invalid: distance > 3000

b. Write the test cases for strong normal equivalence class testing.

(5, 250)

(5,1500)

(20, 250)

(20, 1500)

(40, 250)

(40, 1500)

c. How many test cases would weak normal, weak robust, and strong robust equivalence class testing require? Briefly justify each number.

Weak normal: 3 test cases

Weak robust: 7 test cases

Strong robust: 20 test cases

d. In two or three sentences, compare your test cases from 2(a) and 3(b). What does each set cover well, and what does each miss?

BVT covers the minimum and maximum values well, but it misses the different shipping tiers. Equivalence class testing covers all the tiers, but it does not test the exact boundaries as closely.